

Experiment No-8

Convolution between signals & sequences

Aim: Write the program for convolution between two signals and also between two sequences.

Software Required: Matlab software

Theory:



Convolution involves the following operations.

1. Folding
2. Multiplication
3. Addition
4. Shifting

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \delta(n-k)$$

These operations can be represented by a Mathematical Expression as follows:

$x[n]$ = Input signal Samples

$h[n-k]$ = Impulse response co-efficient.

$y[n]$ = Convolution output.

n = No. of Input samples

h = No. of Impulse response co-efficient.

Example : $X(n) = \{1, 2, -1, 0, 1\}$, $h(n) = \{1, 2, 3, -1\}$

Program:

```
clc;
close all;
clear all;
%program for convolution of two sequences
x=input('enter input sequence: ');
h=input('enter impulse response: ');
y=conv(x,h);
subplot(3,1,1);
stem(x);
xlabel('n');
ylabel('x(n)');
title('input sequence')
subplot(3,1,2);
```

```

stem(h);
xlabel('n');
ylabel('h(n)');
title('impulse response sequence')
subplot(3,1,3);
stem(y);
xlabel('n');
ylabel('y(n)');
title('linear convolution')
disp('linear convolution y=');

```

```

disp(y)
%program for signal convolution
t=0:0.1:10;
x1=sin(2*pi*t);
h1=cos(2*pi*t);
y1=conv(x1,h1);
figure;
subplot(3,1,1);
plot(x1);
xlabel('t');
ylabel('x(t)');
title('input signal')
subplot(3,1,2);
plot(h1);
xlabel('t');
ylabel('h(t)');
title('impulse response')
subplot(3,1,3);
plot(y1);
xlabel('n');
ylabel('y(n)');
title('linear convolution');

```

RESULT: convolution between signals and sequences is computed.

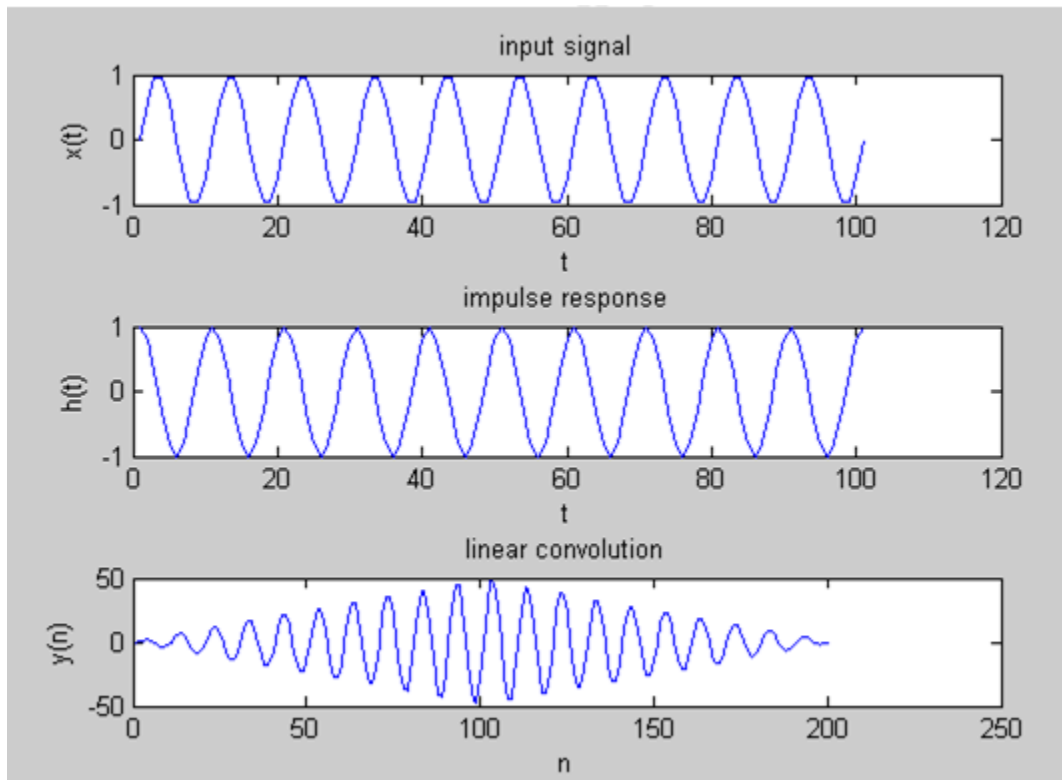
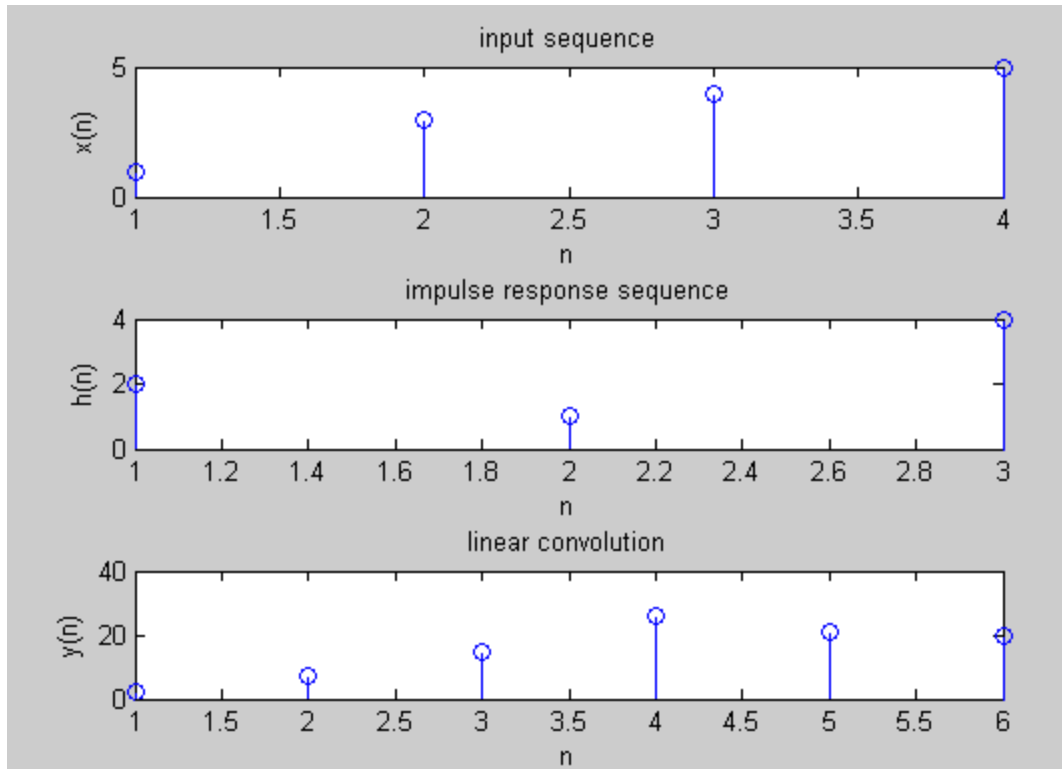
Output:

enter input sequence: [1 3 4 5]

enter impulse response: [2 1 4]

linear convolution y=

2 7 15 26 21 20



VIVA QUESTIONS:-

1. Define Convolution?
2. Define Properties of Convolution?
3. What is the Difference between Convolution & Correlation?
4. Define impulse response?
5. What is Half Wave Symmetry?